Customer Shopping Behavior Behavior Analysis

Uncovering spending patterns, customer segments, and product preferences from **3,900 transactions** to guide strategic business decisions.

Dataset at a Glance

3,900	18	37
Transactions	**Features**	**Missing Ratings**

Key Data Dimensions

- **Demographics:** Age, Gender, Location, Subscription Status
- **Purchase Details:** Category, Amount, Season, Size, Color
- **Behavior:** Discounts, Promo Codes, Frequency, Review Rating, Shipping Type

Data Preparation in Python

01

Load & Explore

Imported dataset with `pandas`; used `df.info()` and `.describe()` for structure review.

02

Clean & Standardize

Imputed missing `Review Rating` values with category median; renamed columns to snake_case.

03

Feature Engineering

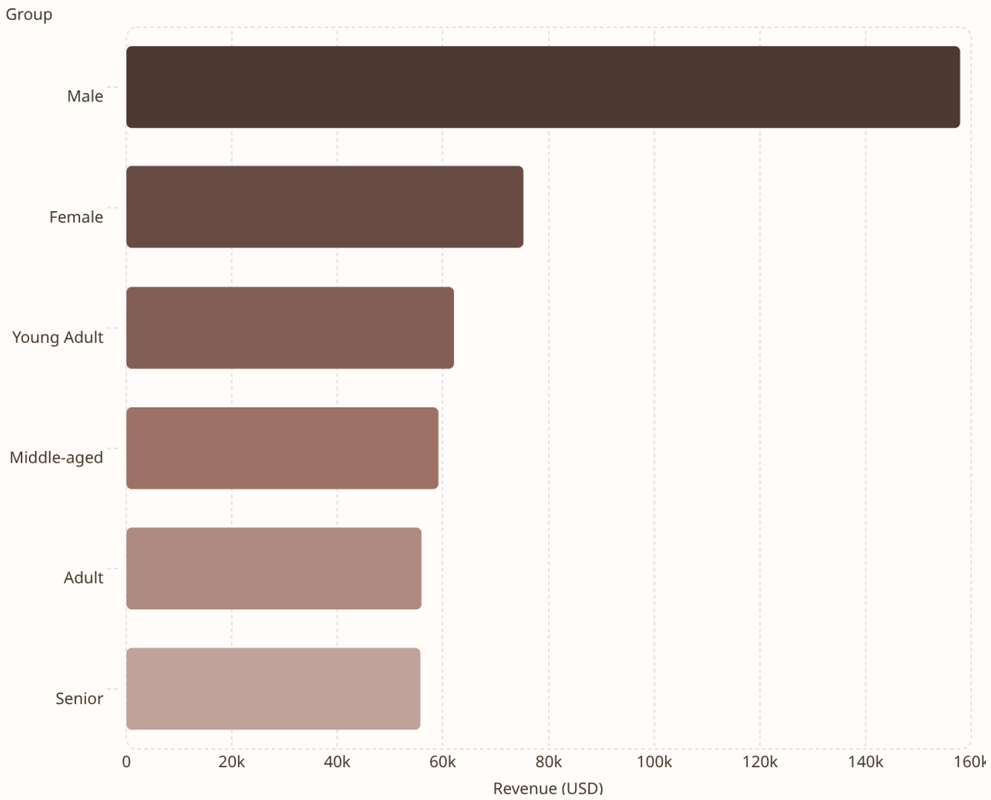
Created `age_group` and `purchase_frequency_days` columns; dropped redundant `promo_code_used`.

04

Database Integration

Loaded cleaned DataFrame into **PostgreSQL** for structured SQL analysis.

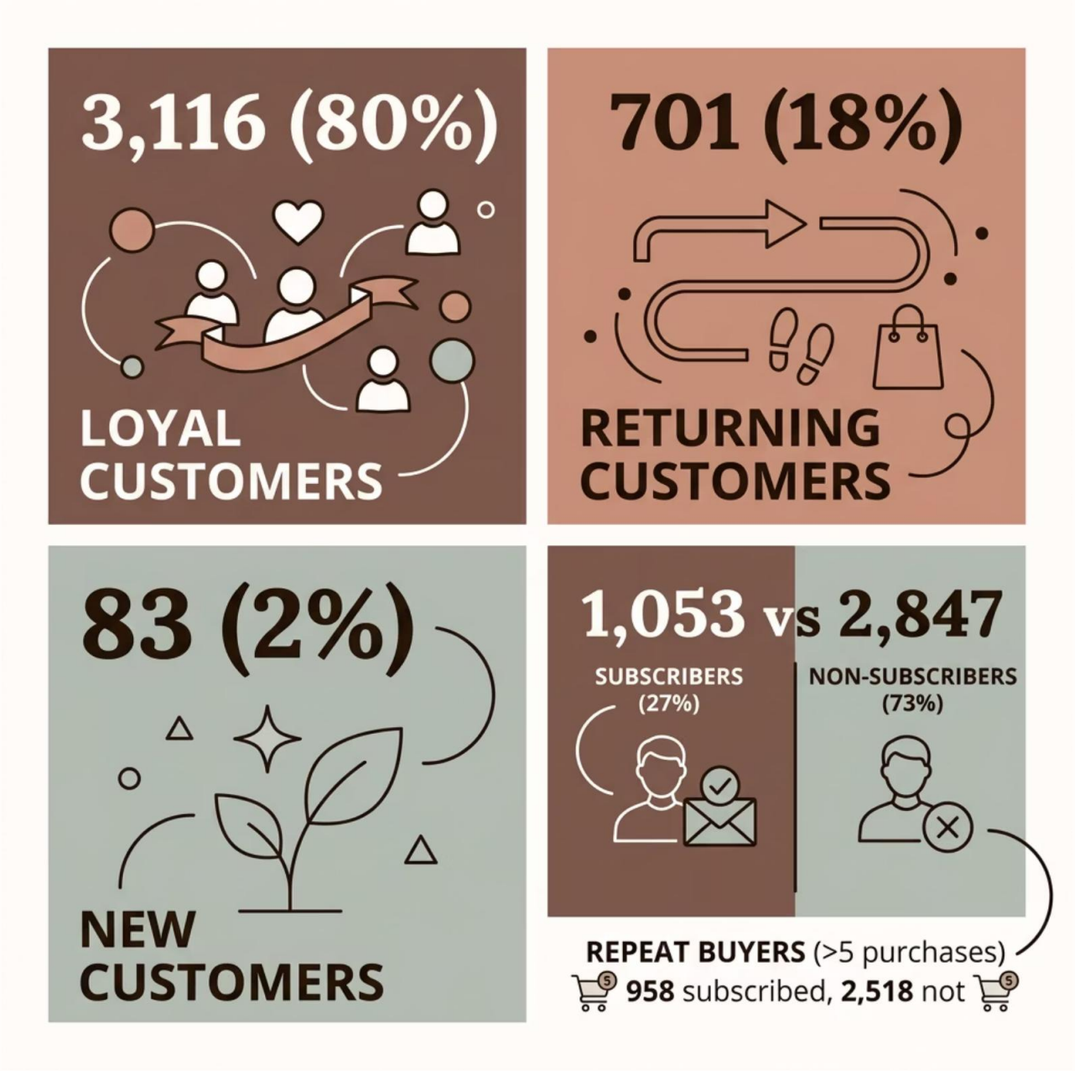
Revenue & Spending Insights



Key Findings

- Male customers generated **$157,890** vs. female customers' **$75,191**
- Young Adults lead revenue by age group at **$62,143**
- Express shipping users spend slightly more (**$60.48**) vs. Standard (**$58.46**)
- **839 customers** used discounts yet still spent above average

Subscription & Loyalty Analysis



Subscription Status

- **27%** subscribers (1,053) vs. **73%** non-subscribers (2,847)
- Avg. spend nearly equal: subscribers **$59.49**, non-subscribers **$59.87**

Customer Segments

- **Loyal:** 3,116 customers
- **Returning:** 701 customers
- **New:** 83 customers

Repeat buyers (>5 purchases) skew non-subscribed — a key conversion opportunity.

Top Products & Discount Behavior

Top 5 Products by Rating

Product	Avg. Rating
Gloves	3.86
Sandals	3.84
Boots	3.82
Hat	3.80
Skirt	3.78

Most Discount-Dependent Products

Product	Discount Rate
Hat	50.00%
Sneakers	49.66%
Coat	49.07%
Sweater	48.17%
Pants	47.37%

Top Products by Category

👗 Clothing

- Blouse — 171
- Pants — 171
- Shirt — 169

💍 Accessories

- Jewelry — 171
- Sunglasses — 161
- Belt — 161

👟 Footwear

- Sandals — 160
- Shoes — 150
- Sneakers — 145

🧥 Outerwear

- Jacket — 163
- Coat — 161

Customer Behavior Dashboard

Subscription Status

No | Yes

Gender

Female | Male

Category

- Accessories
- Clothing
- Footwear
- Outerwear

Shipping Type

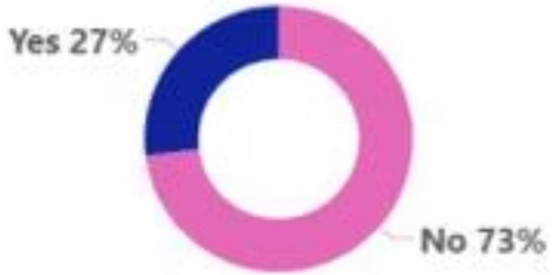
- 2-Day Shipping
- Express
- Free Shipping
- Next Day Air
- Standard
- Store Pickup

3.9K
Number of Customers

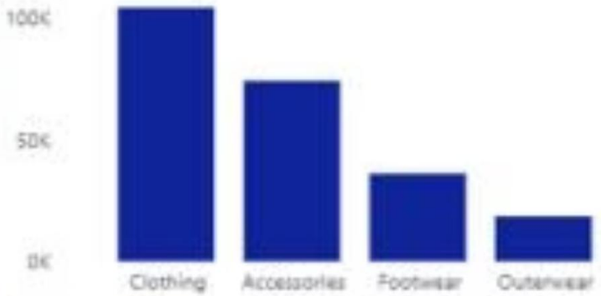
$59.76
Average Purchase Amount

3.75
Average Review Rating

% of Customers by Subscription Status



Revenue by Category

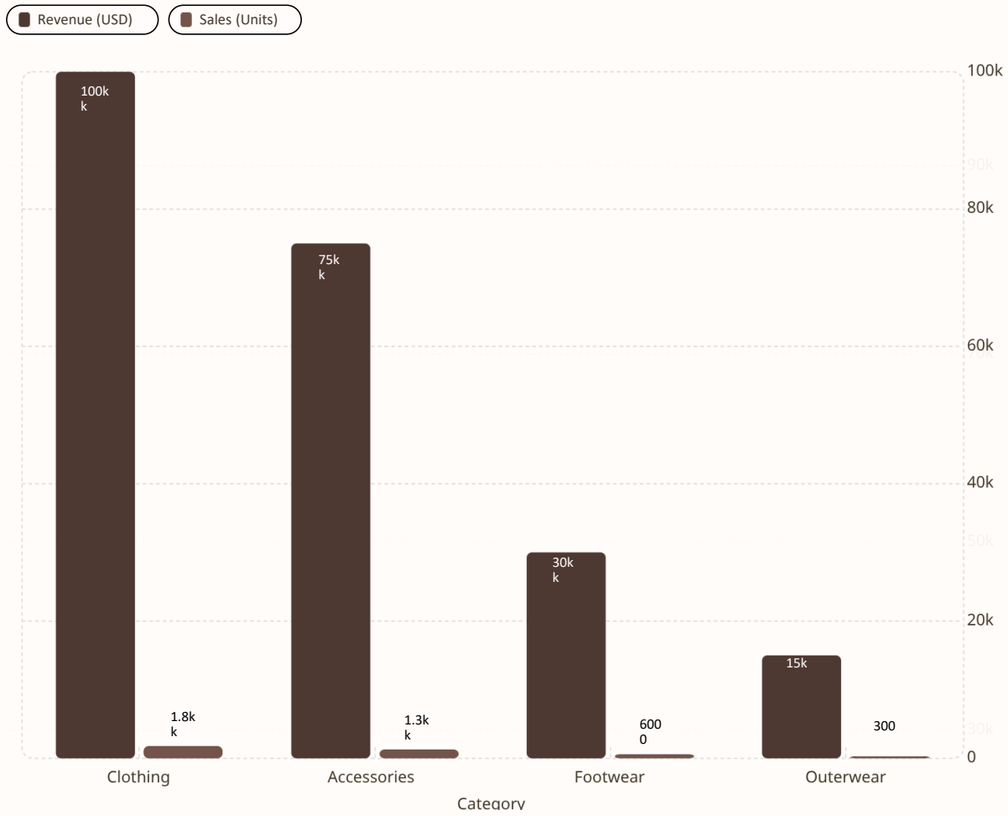


Sales by Category

Revenue by Age Group

Sales by Age Group

Revenue by Category



Category Highlights

- **Clothing** dominates with ~$100K revenue and ~1,800 units sold
- **Accessories** strong at ~$75K revenue
- **Footwear** and **Outerwear** are growth opportunities

Clothing alone accounts for the majority of total revenue — a key focus area for campaigns.

Business Recommendations

Boost Subscriptions

Promote exclusive benefits to convert the 73% non-subscriber base.

Loyalty Programs

Reward repeat buyers to accelerate movement into the "Loyal" segment.

Review Discount Policy

Balance sales boosts with margin control — especially for Hat, Sneakers, and Coat.

Targeted Marketing

Focus on high-revenue age groups (Young Adults) and express-shipping users.

Product Positioning

Highlight top-rated products (Gloves, Sandals, Boots) in promotional campaigns.